Fiscal Year:	FY 2007	**Task Last Updated:**	FY 01/15/2008
PI Name:	Sivathanu, Yudaya R. Ph.D.		
Project Title:	SMALL ANGLE SCATTERING FOR CHARACTERIZATION OF COMBUSTION GENERATED PARTICULATE		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47906	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Feikema, Douglas A	**Contact Phone:**	216-433-5707
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2007 per PI (11/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feikema, Douglas (NASA)		
Grant/Contract No.:	NNC04AA40A		
Performance Goal No.:			
Performance Goal Text:			
	There are four specific tasks associated with achieving the technical objectives of the project. The first task is to design, fabricate, and calibrate a small angle multi-wavelength scattering instrument. The second task is to develop an algorithm to obtain particle size distribution, number density, refractive indices, and fractal dimension from the measured intensities. The third task is to validate the measurement and data reduction scheme using well characterized normal gravity and reduced gravity fires. The last task is to obtain signatures from a variety of particulate, both from combustion of common materials as well as from nuisance sources. These tasks will form the basis of a science requirement document proposing the instrument be evaluated in the Space Station CIR module. The project was limited to three years and therefore reduced gravity data was not obtained.		

Task Description:	There are two key objectives to the proposed work. (1) The first is to utilize advances in measurement and data interpretation technologies to develop a small angle scattering system that can be used in combustion science studies. (2) The second is a systematic study of the scattering signatures of a wide variety of combustion generated particulate and particulate that are present in the environment with a view of augmenting current fire safety systems utilized in the space station. It is critical that advanced optical instrumentation be developed for microgravity combustion research on the ISS. The proposed instrument, which enables characterization of combustion generated particulate, holds great promise to be widely used and is an important instrument for space-based research. An additional benefit of the project is that the fundamental science for developing false alarm free fire detection systems for spacecraft will be completed. The project involves developing an entire system that is flight worthy, including hardware and analysis software, and delivering the system to NASA for use by other researchers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The small angle scattering sensor will form the basis for an advanced, low false alarm rate fire detector system. Such low false alarm rate fire detector systems will have immediate acceptance in the protection of high value (museums, bank vaults, etc.) and confined (interior of submarines, airplanes, etc) spaces.
Task Progress:	During the first year of the project, the design of a bench top small angle scattering system was completed. Components of the system were purchased and assembled on an optical bench. The system was calibrated with 10 mm single slit. During the first year the system was calibrated with a 514 nm TE cooled diode laser. Based on the calibration, it was ascertained that the system can provide accurate data from 0.05 degrees to 10.5 degrees with an increment of less than 0.1 degrees. During the second year, the design and fabrication of the two wavelength small angle scattering system (SASS). The system was designed using ProEngineer solid model package. The optical design was completed using Zemax Optical Design software. The system was calibrated using a 10 micron slit. Scattering signatures were obtained from burning PVC (electrical wire), rubber, and paper. Simulations of the expected scattering from soot particles were also calculated using the The Rayleigh-Debye-Gans Fractal Aggregate (RDG-FA) theory. During the final year of the project, additional evaluation of the system was completed. It was found that the optical design had some problems with not providing correct signatures when the particles are not within a narrow range of distances from the imaging lenses. Attempts to correct this problem were not successful. A deconvolution algorithm was developed during the final year of the project and tested with synthetic data. The data collected from the burning samples were not deconvoluted due to the uncertainties arising identifying the particle. Additional resources are required to correct the optics for the scattering system before it can be utilized with confidence.
Bibliography Type:	Description: (Last Updated: 01/07/2020)